Phishing Email Detection & Awareness Report

Cybersecurity Internship Task 02

Prepared For : Future Interns Cybersecurity Internship Program

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Executive Summary

What is Phishing?

Phishing is currently one of the most common and costly cyber-attacks faced by businesses today. This attack is a technique where attacker impersonate trusted entities (such as a bank or a company) to deceive users.

Why is it dangerous?

Attackers trick users into participating in the following actions:

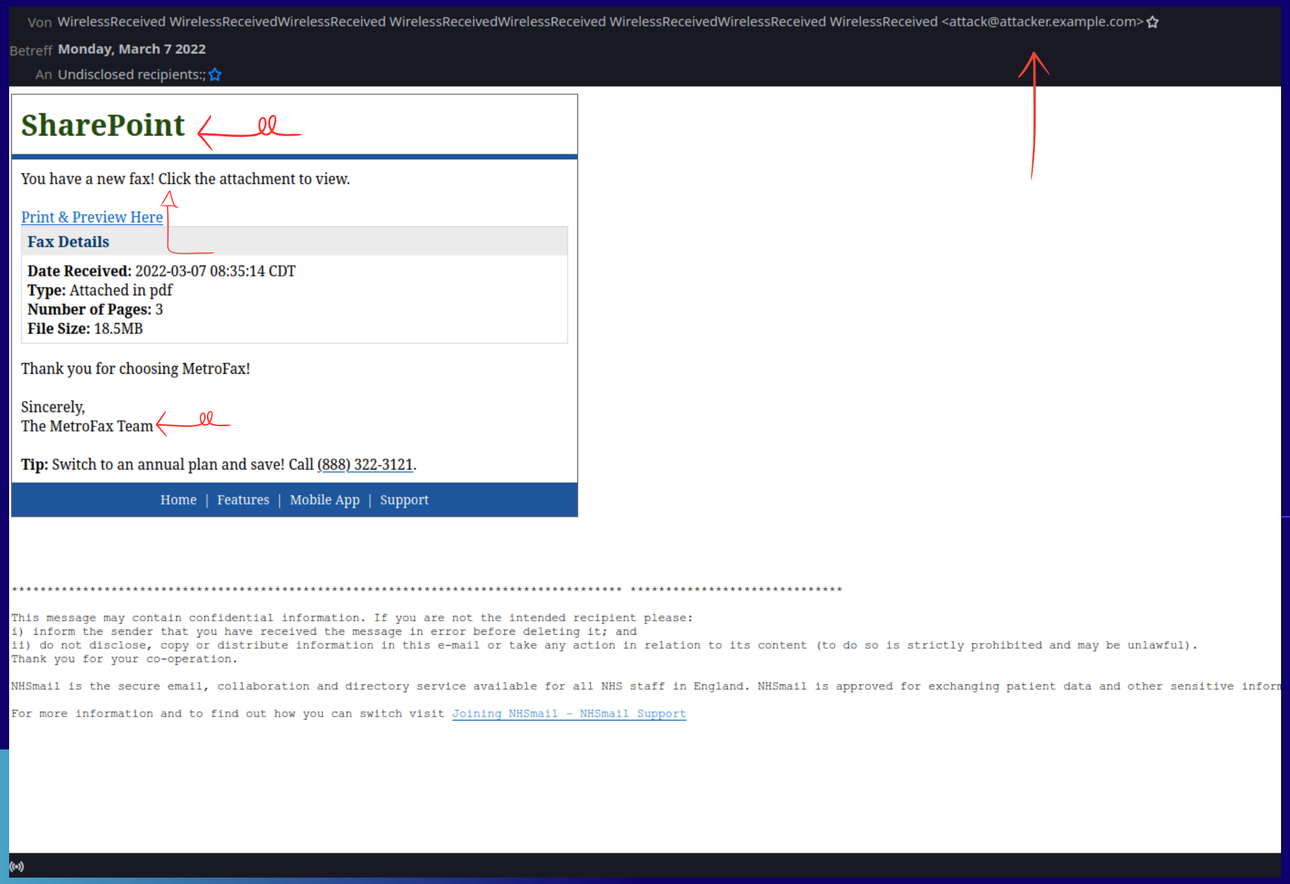
- Clicking malicious links.
- Downloading infected files.
- Sharing passwords or OTPs.

Objective of this Report

This report is designed to educate employees by analyzing the following key points:

- Analyzing real phishing email samples.
- Identifying common phishing indicators.
- Classifying email risk clearly.
- Providing prevention and awareness guidelines to help users avoid phishing attacks.

Phishing Email Analysis (Case Study 1)



Von WirelessReceived WirelessReceivedWirelessReceived WirelessReceivedWirelessReceived WirelessReceivedWirelessReceived WirelessReceived <attack@attacker.example.com>

Betreff **Monday, March 7 2022**

An Undisclosed recipients:;

SharePoint

You have a new fax! Click the attachment to view.

Print & Preview Here

Fax Details

Date Received: 2022-03-07 08:35:14 CDT
Type: Attached in pdf
Number of Pages: 3
File Size: 18.5MB

Thank you for choosing MetroFax!

Sincerely,
The MetroFax Team

Tip: Switch to an annual plan and save! Call (888) 322-3121.

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This message may contain confidential information. If you are not the intended recipient please:
i) inform the sender that you have received the message in error before deleting it; and
ii) do not disclose, copy or distribute information in this e-mail or take any action in relation to its content (to do so is strictly prohibited and may be unlawful).
Thank you for your co-operation.

NHSmail is the secure email, collaboration and directory service available for all NHS staff in England. NHSmail is approved for exchanging patient data and other sensitive inform

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Findings (Red Flags Identified):

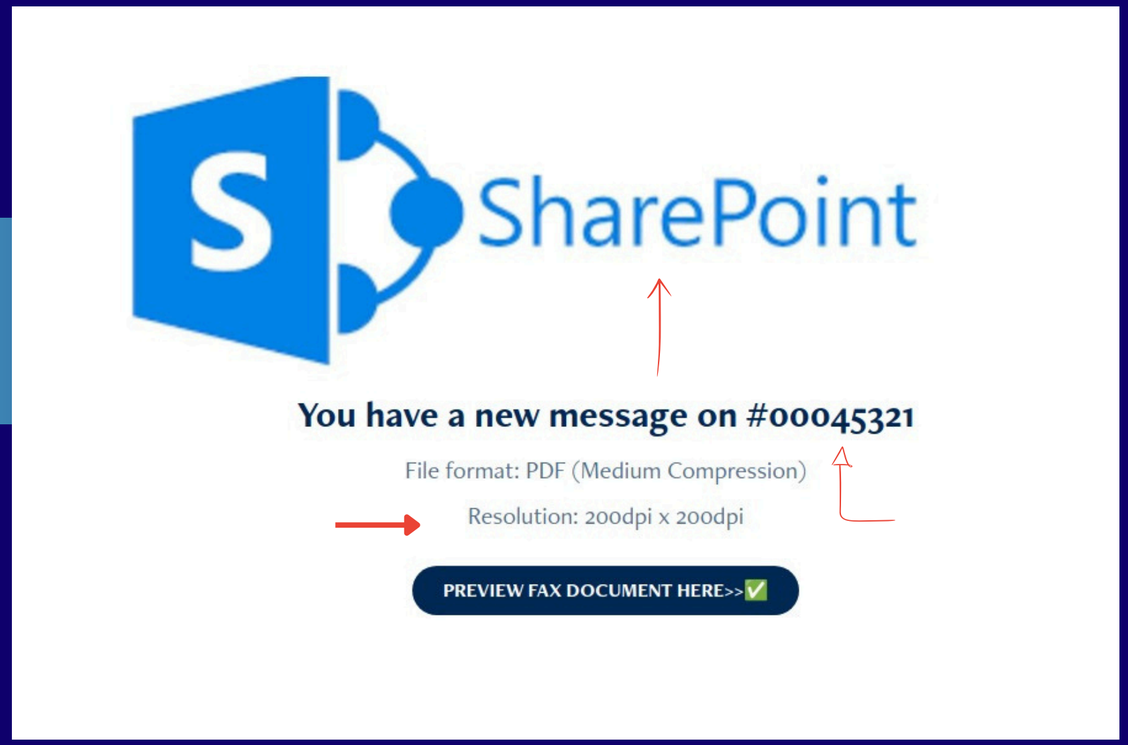
- Suspicious Sender: The sender's email domain does not align with the legitimate organizational email address. This is a primary indicator of identity impersonation.
- False Urgency: The content uses aggressive, fear-based language (e.g., "You have a new fax!") to manipulate the recipient into reacting without critical thought.
- Generic Context: The mixture of "SharePoint" and "MetroFax" in a single email is inconsistent and indicates a lack of professional branding, which is a clear sign of a template-based phishing attempt.

Risk Classification

Indicator	Risk Level
Suspicious Sender	HIGH
Urgency Language	MEDIUM
Generic Context:	MEDIUM

Overall Risk Assessment: HIGH

Phishing Email Analysis (Case Study 2)



Analysis Findings (Red Flags)

Upon analyzing the email, the following phishing indicators were identified:

- **Inconsistent Service Branding:** The email claims to be from "SharePoint" (a Microsoft document management tool),but informs the user about a "Fax" document.SharePoint is not typically used for faxing, which is a major red flag indicating a fake notification. **(Risk Level:HIGH)**

- **Use of Fake Technical Jargon:** To appear legitimate, the attackers used technical-sounding details like "PDF (Medium Compression)" and "Resolution: 200dpi x 200dpi." This is a social engineering tactic intended to trick the user intobelieving the file is a genuine scanned document. **(Risk Leve:MEDIUM)**

- **Vague Notification:** The message uses a generic ID ("#00045321") rather than providing specific, verifiable context (like sender name or document name), which is common in automated phishing templates. **(Risk Leve:MEDIUM)**

Prevention & Awareness Guidelines

- **Stay Safe: The "Do's and Don'ts"**

To maintain a strong defense against phishing, all employees should follow these essential guidelines:

Do's

- ✓ **Verify:** Always check the sender's actual email address.
- ✓ **Hover:** Hover over buttons/links to see the real destination.
- ✓ **Report:** Forward suspicious emails to the IT Security Team.

Don'ts

- ✗ **Don't Click:** Never click links in unexpected emails.
- ✗ **Don't Download:** Avoid opening attachments from untrusted sources.
- ✗ **Don't Share:** Never provide passwords or OTPs via email.

Incident Response

What to do if you receive a suspicious email?

If you suspect an email is a phishing attempt, follow these steps immediately:

1. **Stop & Assess**: Do not interact with the content (no clicks, no downloads).
2. **Report:** Use the "Report Phishing" button in your email client or forward the email to [Insert IT Security Email Address].
3. **Delete**: Once reported, permanently delete the email from your inbox and " Deleted Items" folder.
4. **Confirm**: If you are unsure, contact the IT department through an official, trusted channel (like an internal chat or phone call) before taking any action.

Conclusion

Security is Everyone's Responsibility

Phishing is a human-targeted attack. Technology alone cannot stop it. By staying vigilant, questioning unexpected emails, and following these reporting protocols, we can collectively secure our organizational data.

When in doubt, always report it!